

Graph neural networks for electroencephalogram analysis: Alzheimer's disease and epilepsy use cases

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ABSTRACT

Electroencephalography (EEG) is widely used as a non-invasive technique for the diagnosis of several brain disorders, including Alzheimer's disease and epilepsy. Until recently, diseases have been identified over EEG readings by human experts, which may not only be specific and difficult to find, but are also subject to human error. Despite the recent emergence of machine learning methods for the interpretation of EEGs, most approaches are not capable of capturing the underlying arbitrary non-Euclidean relations between signals in the different regions of the human brain. In this context, Graph Neural Networks (GNNs) have gained attention for their ability to effectively analyze complex relationships within different types of graph-structured data. This includes EEGs, a use case still relatively unexplored. In this paper, we aim to bridge this gap by presenting a study that applies GNNs for the EEG-based detection of Alzheimer's disease and discrimination of two different types of seizures. To this end, we demonstrate the value of GNNs by showing that a single GNN architecture can achieve state-of-the-art performance in both use cases. Through design space explorations and explainability analysis, we develop a graph-based transformer that achieves cross-validated accuracies over 89% and 96% in the ternary classification variants of Alzheimer's disease and epilepsy use cases, respectively, matching the intuitions drawn by expert neurologists. We also argue about the computational efficiency, generalizability and potential for real-time operation of GNNs for EEGs, positioning them as a valuable tool for classifying various neurological pathologies and opening up new prospects for research and clinical practice.

1. Introduction

Electroencephalography (EEG) has been used traditionally to study brain activity and has shown potential in assisting the diagnosis of several brain disorders (Da Silva, 2023) or as an enabler of brain-machine interfaces (Gu et al., 2021). EEG is able to capture the temporal dynamics of neuronal firing over multiple channels distributed spatially over the patient's head, therefore providing spatio-temporal information about simple and complex brain activity.

Among the different applications of EEGs, two key clinical use cases are the diagnosis of Alzheimer's Disease (AD) (Cassani, Estarellas, San-Martin, Fraga, & Falk, 2018) and seizure analysis and discrimination (Faiman, Smith, Hodsoll, Young, & Shotbolt, 2021) since, in both problems, disorders can produce alterations in the EEG patterns. Given the relatively high prevalence and progressive neurodegenerative impact of AD, as well as the grave effects of wrongly medicating misdiagnosed seizures, EEG therefore emerges as a very relevant technique for both use cases.

In the diagnosis of AD, brain connectivity alterations during the evolution of the disease can be a distinctive biomarker, and thus EEG can be employed to distinguish between healthy patients and patients with mild cognitive impairments or more advanced AD (Cassani et al., 2018). Regarding seizure discrimination, people affected by Psychogenic Non Epileptic Seizures (PNES) may show the same paroxysmal events as people affected by epilepsy but they do not exhibit any ictal pattern during the critical event. Also, resting-state EEG can be used to discern whether seizures of a given patient can be attributed to an epileptic or psychogenic origin (Faiman et al., 2021).

Until recently, diseases have been identified over EEG readings by human experts, which may not only be specific and difficult to find, but are also subject to human error, which might be problematic due to the delicate nature of the disorders being treated. On the one hand, in the case of AD, one may want to identify the disease in its early stages, where its symptoms appear in the form of subtle and

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mild cognitive impairments. On the other hand, the difference between epileptic and psychogenic non-epileptic seizures has been explored less in the literature and can be also non-trivial to identify.

As a result, machine learning methods have been applied over the years in an attempt to deliver a superhuman ability in detecting certain specific patterns from the EEGs, and hence provide robust and reliable diagnosis of such diseases. In particular, there are lots of works using techniques such as Support Vector Machine (SVM) (Ahmadi, Pei, Carrette, Aldenkamp, & Pechenizkiy, 2020; Zhao et al., 2021), Multi-Layer Perceptron (MLP) (Ieracitano, Mammone, Hussain, & Morabito, 2020), Long Short-Term Memory (LSTM) (Khademi, Ebrahimi, & Kordy, 2022) or Convolutional Neural Network (CNN) (Ieracitano, Mammone, Bramanti, Hussain, & Morabito, 2019; Giudice, Ferlazzo, et al., 2022; Morabito et al., 2016) applied to problems not limited to AD diagnosis (Cassani et al., 2018) or seizure discrimination (Faiman et al., 2021), but also to motor imagery (Khademi et al., 2022) and developing better brain-computer interfaces (Aggarwal & Chugh, 2022). Despite the positive results presented in the literature, it is argued that these approaches are not capable of capturing the underlying arbitrary non-Euclidean relations between signals in the different regions of the brain. For instance, while CNNs have been used frequently in EEG classification, they imply compressing the relational data of the EEG into a pseudo-image. Since the different parts of the brain are not necessarily correlated by distance, CNNs hence appear to be sub-optimal.

In this context, Graph Neural Networks (GNNs) have gained attention for their ability to effectively analyze complex relationships within different types of data (Bai et al., 2024; Li, Micheli, et al., 2024; Li, Zhou, Chen, Huang & Jiang, 2024; Wu et al., 2020; Wu, Sun, Zhang, Xie, & Cui, 2022; Xu, Hu, Leskovec, & Jegelka, 2019; Zhou et al., 2020), including EEGs (Hou et al., 2022; Klepl, He, Wu, Blackburn, & Sarrigiannis, 2022; Rasheed et al., 2021; Wang et al., 2020). Indeed, EEGs can be represented as a graph where each channel is seen as a node, and the possible interactions between these nodes, represented as edges, are seen as the activity of the different regions of the brain. Therefore, making use of geometric graph-based deep learning methods such as GNNs provides a more suitable approach to learn and leverage the valuable information present not only in the nodes' features, but also in the connectivity patterns of EEG data.

Despite being uniquely suited to EEG classification tasks, the number of works employing GNN methods is relatively limited (Rasheed et al., 2021). As further developed in Section 3, few works have employed simple GNN variants in the context of AD diagnosis (Klepl et al., 2022), epilepsy detection (Wang et al., 2020) or motor imagery (Hou et al., 2022).

The results obtained therein are important as they improve the accuracy over prior works, yet we argue that the potential of GNNs is not fulfilled completely due to the lack of a design space exploration justifying the employed techniques. Moreover, these existing works are limited to a single use case, often with binary classification only. Further, they do not exploit the inherent explainability offered by GNNs, which allow to evaluate the features, edges, and nodes that have a higher relevance during inference.

To bridge this gap, this paper aims to demonstrate the unique suitability of GNNs in EEG classification tasks through two relevant use cases: AD diagnosis and epileptic seizure discrimination (Fig. 1). In this direction, the main contributions of the work can be summarized as follows:

- We perform a design space exploration of different GNN architectures, including less explored but powerful GNN transformers, to find GNN architectures that maximize the accuracy of the proposed AD and seizure classification tasks while being computationally efficient.
- We provide a cross-validated assessment of both AD and seizure discrimination tasks using a *single* GNN architecture that outperforms or performs close to the state of the art in *both* use cases simultaneously.

- We evaluate the explainability of the proposed architecture through node, edge, and feature importance analyses to understand whether the patterns learnt by the GNN match expert knowledge in both use cases.
- We finally provide a qualitative discussion over the advantages of GNNs beyond accuracy and explainability, including aspects such as generalizability, scalability, and potential use for real-time applications.

The proposed architecture is based on GNN transformers and achieves over 89% and 96% in the ternary classifications of AD diagnosis and seizure discrimination tasks, achieving a remarkable improvement over the state of the art. The models are cross-validated through a leave-one-out method, and their explainability assessed through feature, node, and edge importance analyses using GNNExplainer (Ying, Bourgeois, You, Zitnik, & Leskovec, 2019).

The remainder of the paper is as follows. In Section 2, background on the topic of EEGs is provided, along with some fundamental aspects related to the processing of the EEG signals, and more details on the AD and epilepsy use cases. Then, in Section 3, the literature on machine learning applied to the aforementioned use cases is surveyed. In Section 4, the materials and methods of are explained, including details of the datasets used, the preprocessing methods, the evaluated GNN architectures, and the validation and explainability techniques used. Subsequently, in Section 5, we present the results of the two use case evaluations, including design space explorations and a comparative evaluation with the state of the art. Finally, the results are discussed from the prism of explainability in Section 6 and the paper is concluded in Section 7.

2. Background

2.1. Electroencephalograms

EEG is a non-invasive technique commonly used in clinical settings to diagnose and monitor various neurological conditions, including epilepsy (Rosenow, Klein, & Hamer, 2015; Tatum et al., 2018), sleep disorders (Aboalayon, Faezipour, Almuhammadi, & Moslehpour, 2016; Heyat, 2017), brain tumors (Selvam & Devi, 2015), and brain injuries (Curley, Forgacs, Voss, Conte, & Schiff, 2018). It can also be used in research settings to study brain activity during different tasks or cognitive processes (Chikhi, Matton, & Blanchet, 2022; Kuanar, Athitsos, Pradhan, Mishra, & Rao, 2018). It records and analyzes the electrical activity of the brain by placement of electrodes on the scalp to measure and record the electrical signals generated by neurons in the brain with excellent temporal resolution. Patients undergoing EEG recording have no complications or discomfort, which is why this technique is widely tolerated.

Typically, electrodes are placed over the scalp according to the International 10–20 system, a standardized method for EEG recording also used in this study (Jasper, 1983). In this system, the numbers “10” and “20” refer to the distances between adjacent electrodes, which are set at 10% or 20% of the total front-back or right-left distance of the head. Fig. 2 illustrates the spatial arrangement and classification of electrodes according to different brain areas. In particular, each electrode is labeled with a letter representing one of the six areas: Pre-frontal (Fp), Frontal (F), Temporal (T), Parietal (P), Occipital (O), and Central (C). Additionally, each electrode is assigned a number, where even numbers indicate electrodes on the right hemisphere and odd numbers indicate electrodes on the left one. Electrodes labeled with the letter “z” indicate midline placement, while those labeled with the letter “a” indicate auricular placement.

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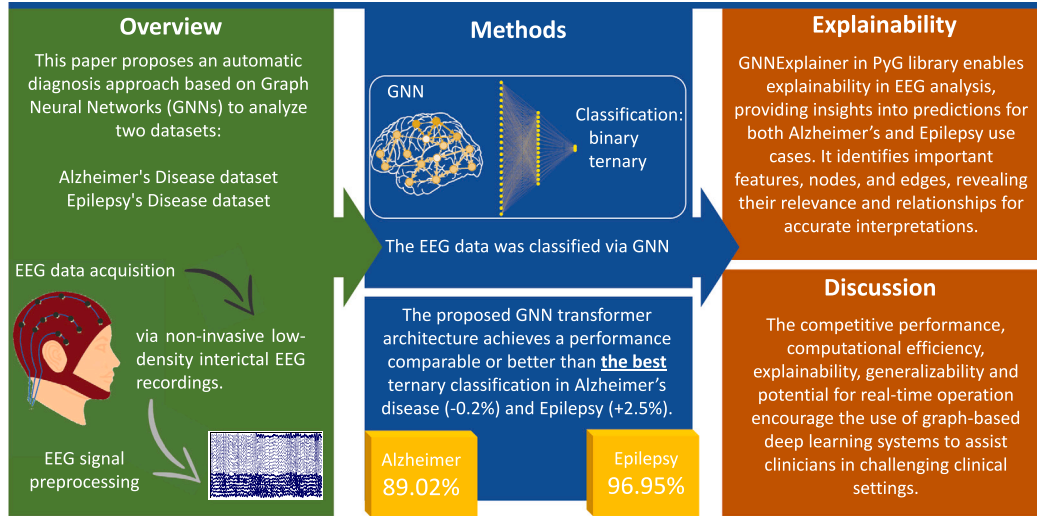


Fig. 1. Graphical abstract of the paper.

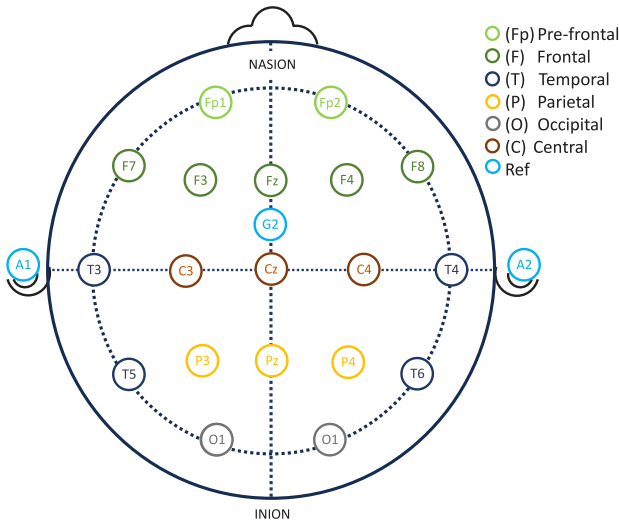


Fig. 2. The standard 10–20 electrode montage used in the study. For Alzheimer's dataset includes references on A1 and A2, while the epilepsy dataset includes a reference on G2.

2.2. Signal processing aspects

Clinical interpretation of EEG recordings requires physicians with specialized knowledge and expertise on visual inspection of the tracing and quantitative EEG analysis. To achieve a deep understanding of the underlying brain activity and to detect latent biomarkers, it is essential to employ advanced analysis techniques involving engineering metrics and signal processing aspects. Features not evident on visual inspection can be grasped only with such techniques. Several examples include metrics such as the Phase-Lag Index (PLI) or the Pearson correlation, which work over the raw signal or spectral transformations of it.

Spectral transformations encompass a range of analysis techniques such as Fourier transform, wavelet transform, and time-frequency analysis methods. These techniques allow researchers to examine the power distribution across different frequency bands within the EEG signal. By employing spectral transformations, researchers can identify specific frequency components associated with various brain states or cognitive processes. Power Spectral Density (PSD) is the most common feature

extraction method used (Ong, Saidatul, & Ibrahim, 2018). The PSD is effective for processing stationary signals and well suitable for narrow-band signals. It distributes signal power over frequency and shows the magnitude of energy power as a function of frequency.

The PLI is a metric that has received growing attention in the field of functional connectivity analysis recently (Chaturvedi et al., 2019; Stam, Nolte, & Daffertshofer, 2007). It facilitates the estimation of phase synchronization between different EEG channels, thereby enabling the identification of functional connectivity patterns and information flow between distinct brain regions. PLI is designed to address the issue of volume conduction in functional connectivity analysis. Indeed, by focusing on the phase information of the EEG signals rather than their amplitudes, the PLI provides a measure that is less affected by the mixing effects of volume conduction.

The Pearson correlation, a widely utilized statistical measure, enables the assessment of the linear relationship between EEG channels. By calculating the Pearson correlation coefficient, it becomes possible to determine the strength and direction of associations between different brain regions based on their raw EEG signal or their associated transformations, providing insights into their functional connectivity.

The utilization of these advanced analysis techniques provides quantitative measures and enables rigorous investigation of EEG data. Such techniques facilitate statistical comparisons and contribute to an improved understanding of brain dynamics, functional connectivity, and the relationships between various brain regions or cognitive processes. However, it is important to acknowledge that visual inspection continues to hold value in conjunction with these quantitative analyses. Visual inspection aids in the identification of artifacts or irregularities that automated algorithms may overlook, while also providing a qualitative assessment of the data.

2.3. Use of AI for EEG analysis

With the exponential growth of Artificial Intelligence (AI), there is potential for advanced algorithms to support the interpretation of EEG data. These algorithms can be trained using large datasets that incorporate knowledge gained from clinicians' annotations on numerous EEG recordings. By leveraging AI techniques, the algorithms can learn patterns and characteristics within the EEG data that are indicative of specific conditions or events. The training process involves exposing the algorithm to a diverse range of annotated EEG recordings, allowing it to recognize and extract relevant features automatically.

Table 1

Summary of related work: AI methods applied to the diagnosis of Alzheimer's disease and the discrimination of epileptic seizures.

Reference	Method	Alzheimer's		Epilepsy		Explainability
		Binary	Ternary	Binary	Ternary	
Zhao et al. (2021)	SVM	✓				
Kim and Kim (2018)	CNN	✓				
Fan, Yang, Fuh, and Chou (2018)	LR	✓				
Morabito et al. (2016)	CNN		✓			
Bi and Wang (2019)	CDBM		✓			
Ieracitano et al. (2019)	CNN		✓			
Ieracitano et al. (2020)	MLP		✓			
Klepl et al. (2022)	GNN	✓				
Giudice, Varone, et al. (2022)	CNN			✓		
Ahmadi et al. (2020)	KNN			✓		
Ahmadi et al. (2020)	SVM			✓		
Ahmadi et al. (2020)	R. Forest			✓		
Ahmadi et al. (2020)	D. Tree			✓		
Ahmadi et al. (2020)	GBoost			✓		
Zsom et al. (2019)	XGBoost			✓		
Giudice, Ferlazzo, et al. (2022)	CNN				✓	
Wang et al. (2020)	SGNN			✓		
This work	GNN	✓	✓	✓	✓	✓

The availability of extensive annotated datasets empowers the AI algorithms to make more accurate interpretations and potentially assist clinicians in their analysis, therefore enhancing their diagnostic capacity. While human expertise remains crucial in the interpretation of EEG data, AI algorithms can augment this process by providing additional insights, flagging anomalies, or highlighting patterns that might be overlooked. Moreover, by automating certain aspects of the interpretation process, AI algorithms can save time, improve efficiency, and hence facilitate better patient care.

It is important to note that the integration of AI into EEG analysis is an ongoing area of research and development, as we describe in subsequent sections. Collaboration between AI specialists and medical professionals is crucial to ensure the algorithms are trained and validated effectively, and the generated insights are clinically relevant and reliable.

3. Related work

There are multiple recent studies on the classification of brain conditions based on EEG signals employing AI techniques. Among these, particular emphasis is placed on AD diagnosis and seizure discrimination as a result of their clinical relevance and impact. A large part of these studies is based on feature extraction methods and various pipelines of different complexity. As a summary, Table 1 shows a representation of the state of the art in the application of AI for EEG-based classification of these two specific conditions, which we explain next.

3.1. Related work in Alzheimer's Disease

Several studies have reported on the topic of AD diagnosis, providing valuable insights and findings. For example, Vialatte et al. (2005) conducted a transformation on a dataset comprising 60 EEG recordings from 22 individuals with Mild Cognitive Impairment (MCI) and 38 Healthy Controls (HC). This transformation involved the use of Morlet's complex wavelet function to generate time-frequency representations. Next, a bump modeling approach was used to generate relevant features from these representations. These selected features were then used as input to an MLP classifier. The results obtained from this classification approach demonstrated outstanding performance, with 93% accuracy, 91% sensitivity, and 94.7% specificity.

Lehmann et al. (2007) conducted a comprehensive investigation by analyzing various aspects of EEG recordings from a sample consisting of 45 HC, 116 individuals with mild AD, and 81 individuals with moderate AD. The measurements included relative and absolute spectral power,

power spectral distribution, and spatial synchronization. Different classification techniques were employed to distinguish between the groups. Specifically, a random forest-based classifier achieved a sensitivity of 85% and specificity of 78% in classifying mild AD versus HC. In the case of moderate AD versus HC classification, SVM and MLP-based classifiers achieved a sensitivity of 89% and specificity of 88%.

In a separate study conducted by McBride et al. (2014), regional spectral and complexity-based features extracted from EEG recordings of 15 HC, 16 individuals with early MCI, and 17 individuals with early AD were evaluated. A three-way SVM-based classifier was proposed and achieved classification accuracy of 83.3%, 85.4%, and 79.2% for the eyes-opened resting state, eyes-closed counting task, and eyes-closed resting state, respectively. These findings contribute to the understanding of the distinguishing characteristics of EEG patterns in different cognitive states.

Ieracitano et al. (2020) presented a novel multi-modal ML-based approach for integrating engineered features from EEG signals to classify brain states in patients with MCI or AD compared to HC subjects. The proposed approach utilizes the Continuous Wavelet Transform (CWT) to project 19-channel EEG signals into the time-frequency (TF) domain and extracts CWT features from EEG sub-bands. Additionally, they extracted higher-order statistics (HOS) from the bispectrum (BiS) representation to capture nonlinear phase-coupling information, resulting in BiS features, that are then used as input to various ML classifiers for 2-way and 3-way classifications. Experimental results using an EEG dataset of 63 AD, 63 MCI, and 63 HC subjects demonstrated that the multi-modal concatenation of CWT and BiS features achieved superior performance with the MLP classifier outperforming other models such as Autoencoder (AE), Logistic Regression (LR), and SVM. The same research team also proposes a data-driven machine deep learning approach, analyzing the PSD of the EEG signals and transforming the spectral profiles into 2D grayscale images called PSD-images (Ieracitano et al., 2019). A customized CNN architecture, trained from the PSD-images, performs binary and three-way classification tasks. Results of the CNN model surpassed conventional learning machines, with an average accuracy of 89.8% in binary classification and of 83.3% in ternary classification.

Fouladi, Safaei, Mammone, Ghaderi, and Ebadi (2022) proposes two deep learning architectures, modified CNN and convolutional autoencoder neural networks (Conv-AE), for classifying subjects with AD, MCI, and HC based on scalp EEG recordings. The EEG data from 61 HC, 56 MCI, and 63 AD subjects were analyzed using TF representation with CWT and Mexican hat function. The modified CNN achieved an average accuracy of 92%, while the Conv-AE network achieved 89% accuracy. These deep learning architectures outperformed conventional machine

learning methods, providing higher accuracy and handling multi-class classification.

Finally, Klepl et al. (2022) uses a Graph Convolutional Network (GCN) architecture using the power spectral densities as node features and try multiple metrics as the weights in the edges of the connectivity graph. The use of weighted PLI and amplitude envelope correlation yield reasonably good results, with a maximum accuracy of 92% only for the binary classification case. Our work differs from this paper in the evaluation that considers both binary and ternary classification cases, as well as in the use of more advanced techniques such as graph-based transformers that learn the correlations leading to a higher accuracy.

3.2. Related work in seizure discrimination

Multiple studies have been conducted regarding epilepsy using EEG signals. For example, Yuan, Zhou, Li, and Cai (2011) presented a Epileptic EEG classification approach using nonlinear dynamical features in a MLP. Nonlinear features, including approximate entropy, Hurst exponent, and scaling exponent, were extracted from EEG signals. Experimental results showed that the proposed model outperformed backpropagation and SVM methods in terms of training time and classification accuracy, achieving an accuracy of 96.5% for differentiating interictal and ictal EEG signals.

In relation to the classification between two types of disorders, recently, Ahmadi et al. (2020) focused on the classification of epilepsy and PNES using short-term EEG data. Different features including signal, functional network, and EEG microstate features were analyzed and demonstrated that the EEG beta sub-band was the most effective for classification. Additionally, the coverage feature of EEG microstate analysis in the beta-band showed high accuracy and precision in classification. Therefore, beta-band and coverage were identified as crucial features for distinguishing between epilepsy and PNES patients.

Pippa et al. (2016) explored different machine learning techniques to classify generalized epileptic and non-epileptic events based on multi-channel EEG data. Signal patterns were extracted and combined to capture spatio-temporal manifestations of seizures. Feature selection improved classification accuracy, with selected feature subsets achieving high accuracies of 95% (Bayesian Network), 89% (Random Committee), and 87% (Random Forest) in binary classification, surpassing previous methods. In a follow-up study (Pippa, Zacharaki, Koutroumanidis, & Megalooikonomou, 2017), the authors also explored different data fusion schemes to combine information from multiple channels, along with dimensionality reduction techniques. The framework was evaluated on EEG data from 11 subjects, achieving better recognition accuracy compared to their previous studies, with a maximum accuracy of 97%.

Giudice, Varone, et al. (2022) differentiated epileptic seizures (ES) from PNES using non-linear EEG analysis and deep learning. They designed a CNN architecture to classify the two categories based on EEG recordings. The CNN achieved 94.4% accuracy, outperforming standard learning algorithms. They also performed the information theoretical analysis, i.e., the permutation entropy, to interpret the CNN performance in order to support neurologists with early diagnoses of ES and PNES. As a follow up to this work, the same group of authors developed a larger CNN model to perform ternary classification between ES, PNES, and HC patients, achieving 85.9% accuracy (Giudice, Ferlazzo, et al., 2022). This is among the very few deep learning models for ternary classification in this particular application.

Finally, the use of GNNs for epilepsy discrimination has not been explored in depth thus far. The work from Wang et al. (2020) is one of the few approaches that use graph learning techniques. However, in their case a sequential GNN (SGNN) is used to transform the time-domain signal of a single electrode into a graph. The evaluation is performed for the binary case and only using data from a single channel. Hence, the results are not comparable to those of this work, which considers a multi-channel architecture.

4. Materials and methods

In this section, we describe the methodology followed in this work, delving into aspects such as the characteristics of the datasets and their pre-processing (Section 4.1), the feature space for the nodes and edges of the input EEG graph (Section 4.2), the considered GNN architecture variants (Section 4.3), as well as the methods for validation (Section 4.4) and for the evaluation of explainability of the GNN models (Section 4.5).

4.1. Datasets

This section describes data acquisition details, patient inclusion and exclusion criteria, and pre-processing procedures for the AD diagnosis and seizure discrimination use cases. A general overview of the methodology is shown in Fig. 3.

Dataset for Alzheimer's Disease Use Case.

The Alzheimer's Disease dataset was collected in previous work for classification purposes Ieracitano et al. (2019). It consists of a total of 189 participants: 63 individuals were diagnosed with AD, 63 individuals were diagnosed with MCI, and 63 participants were included as HC. The participants reflect a diverse demographic, but centered on the average incidence rate of the disease. In this case, AD is more prevalent in older age groups and the dataset reflects that fact.

To meet the inclusion criteria, participants had to be already diagnosed with either AD or MCI (American Psychiatric Association, Association, et al., 2013). The study also established exclusion criteria, which included the presence of other psychiatric or neurological disorders that could potentially cause cognitive impairment, traumatic brain injuries, uncontrolled or complex systemic disorders, and EEG epileptiform patterns.

During the EEG acquisition and recording session, participants were instructed to the full procedure, and sat, with their eyes closed, in a comfortable position ensuring a relaxed state while avoiding falling asleep. This was verified by EEG experts who continuously monitored the patient and EEG traces, promptly identifying any patterns indicative of sleep and other abnormal moments reported with accurate annotation in the accompanying EEG report document. Any EEG segments affected by artifacts or disturbances were identified and marked by the EEG experts for subsequent exclusion from the analysis.

The EEG recordings followed the standard 10–20 International System electrode placement configuration (Jasper, 1983). A 19-channel montage was employed, consisting of electrode placements at Fp1, Fp2, F3, F4, C3, C4, P3, P4, O1, O2, F7, F8, T3, T4, T5, T6, Fz, Cz, and Pz. The EEG signals were sampled at a frequency of 1024 Hz to capture detailed temporal information. To reduce interference from power line frequency, a notch filter was applied at 50 Hz. The linked earlobe (A1-A2) served as the reference for the EEG recordings, ensuring consistency and comparability across channels. Subsequently, the EEG recordings underwent a band-pass filtering procedure ranging from 0.5 to 32 Hz, utilizing the *eegfiltfft* function available in the EEGlab toolbox within MATLAB (Brunner, Delorme, & Makeig, 2013). Subsequently, the recordings were downsampled to a sampling rate of 256 Hz. The data was then divided into five-second non-overlapping epochs. In order to ensure data quality, only EEG segments devoid of artifacts were included. These artifact-free segments were stored on a computer and processed individually, one at a time.

Dataset for Epilepsy Use Case. The Epilepsy's Disease dataset was previously collected in prior works (Giudice, Ferlazzo, et al., 2022; Giudice, Varone, et al., 2022) for the clinical diagnostic procedure of seizure discrimination. The dataset includes 103 subjects analyzed via video-EEG. Among them, 42 individuals were diagnosed with new-onset ES, 42 were diagnosed PNES, and 19 participants were included as HC. The participants reflect a diverse demographic, centered on the average incidence rate of the condition being studied. In this case, the age range is from 11 to 79 years, with the

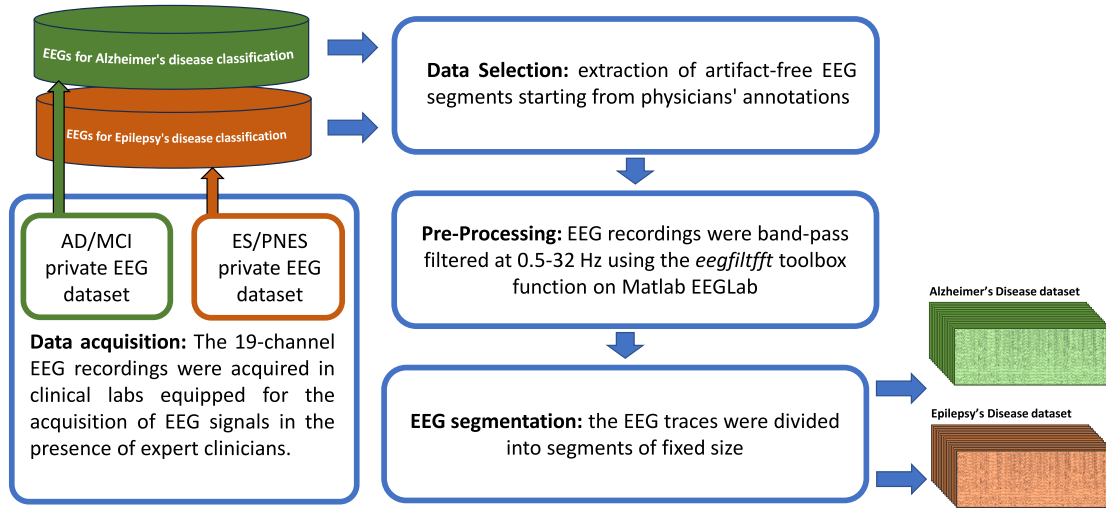


Fig. 3. Dataset acquisition, selection, pre-processing, and segmentation methods considered in this paper.

mean age being 44.69/30.83/33.84 and the standard deviation being 23.11/13.41/13.94 for ES/PNES/HC patients, respectively.

Participants were informed of the diagnostic setup, sat in a quiet, naturally lit room and proceeded with the recording. To meet the inclusion criteria, the intercritical EEG had to be at the visual inspection of experienced clinicians. Participants had to be asked willingness to participate in the study therefore to sign informed consent. An exclusion criterion was chronic use of psychotropic drugs at the time of recording that could have altered the EEG tracing.

The EEG signals were recorded using a 19-channel standard 10–20 system, with the same channel layout than in the AD use case: Fp1, Fp2, F3, F4, C3, C4, P3, P4, O1, O2, F7, F8, T3, T4, T5, T6, Fz, Cz, and Pz. This time, the reference electrode was placed in G2, positioned between electrodes Fz and Cz. To ensure optimal signal quality, the skin impedance values for all electrodes were maintained below 5 K Ω . The EEG data was acquired during a resting condition and recorded for a duration of 20 minutes. Like in the AD use case, the EEG dataset was also subjected to a band-pass filtering procedure ranging from 0.5 to 32 Hz and then downsampled to 256 Hz sampling rate. In this case, however, the data was then divided into in 50% overlapping epochs of 2 seconds. Only EEG segments free of artifacts were included to ensure data quality.

4.2. Graph construction

Relational data can be expressed in multiple ways, leading to multiple graph representations. In the case of EEG data, several node and edge extractors were tested and their performance was evaluated and compared. On the one hand, for **node feature extraction**, we used:

- **Raw data (Raw):** the raw sampled signal values of every channel are used as each node's attributes. This is used generally as a baseline and also to assess the potential of GNNs to lead to an end-to-end solution without intermediate feature extraction.
- **Power Spectral Density (PSD):** given the importance of frequency features in the classification of brain conditions, the average power of a signal for different frequency bands is obtained. In our case, the Gamma band (30 Hz and up), the Higher Beta band (16–30 Hz), the Lower Beta band (12–16 Hz), the Alpha band (8–12 Hz), the Theta band (4–7 Hz) and the Delta band (0–4 Hz) are considered. To obtain the PSD of the data, the Welch method and the Burg method were adopted in this study.
- **Band statistics (Stats):** for each of the aforementioned frequency bands, six statistic descriptors were extracted using standard MATLAB functions: mean, standard deviation, entropy, variance, skewness and kurtosis.

On the other hand, for those GNN architectures that incorporate edge weights, the following **edge feature extractors** were tested:

- **Pearson correlation coefficient (PCC):** uses the covariance and the standard deviation of two signals, as described in Section 2.2, to acquire a value between -1 and 1 where 0 indicates no correlation.
- **Phase Lag Index (PLI):** measurement meant to calculate the similarity between two signals depending on the synchronization between their phases, as described in Section 2.2.
- **Cross spectral density (CSD):** determines how correlated two signals are using the distribution of their PSD.

Finally, in terms of **graph connectivity**, we considered a fully connected graph regardless of the underlying GNN model.

4.3. GNN architectures

In this work, we build multiple GNN architectures based on four different GNN layer variants, namely:

- **Graph Convolutional Networks (GCN):** In this type of architecture, proposed in Kipf and Welling (2016), the features of a given node in each layer are averaged with that of their neighboring nodes, using the edge features as weights. Then, the aggregated features are multiplied by a weight matrix and processed with a non-linear activation function. This whole process is repeated as many times as layers.
- **Graph Attention Networks (GAT):** proposed by Veličković, Cucurull, Casanova, Romero, Liò, and Bengio (2018), GAT is an alternative to the GCN architecture, which provides a multi-headed attention mechanism to calculate edge weights during training using additional trainable parameters. This approach does not require the inclusion of edge weights prior to training, unlike the GCN architecture, but rather allows the model to learn which edges are more relevant.
- **Self-supervised GAT (SelfGAT):** This architecture is a variant of the GAT model designed for noisy graphs and proposed by Kim and Oh (2021). It uses multiple types of attention to better adapt link prediction and edge weighting to different types of graphs.
- **Graph Transformer Networks:** These are an application of the original Transformer architecture (Vaswani et al., 2017) on GNNs which was proposed by Shi, Huang, Wang, Zhong, Feng, and Sun (2020). Similarly to GATs, graph transformers use multi-headed attention to determine the weights of the edges, but in

Table 2
Summary of evaluated GNN architectures.

Short name	Layer type	Layers	Heads
GCN	Graph Convolutional	3	–
GAT	Graph Attention	3	3–4
SelfGAT	Self-supervised Attention	3	3
Transformer	Graph Transformer	3–4	2–8
Transformer (more features)	Graph Transformer	3	3
Transformer (no dropout)	Graph Transformer	3	3
Transformer + GCN	Transformer+Convolutional	4	3

this case the attention mechanism is drawn from the original Transformer architecture (see [Min et al., 2022](#) for more details).

All models have been implemented in PyTorch Geometric ([Fey & Lenssen, 2019](#)) which provides native support for the aforementioned GNN layer types. Then, to determine the best GNN architecture candidates for this work, we performed a guided search to refine basic hyperparameters such as number of layers, hidden dimensions, or activation functions. The best architectures for each type of layer are outlined in [Table 2](#) and described in detail next.

- **GCN:** 3 graph convolutional layers with 32, 32 and 64 features, respectively, which are aggregated using a global addition pooling layer and fed to an MLP. This MLP is formed by 3 fully connected layers with 64, 32 and N nodes, respectively, where N stands for the number of channels.
- **GAT:** 3 graph attention layers with 16, 32 and 64 features, respectively, ending with a dropout layer with a dropout probability of 20%. The dropout layer's output is aggregated using a global addition pooling layer and fed to an MLP. This MLP is formed by 3 fully connected layers with 192, 96 and N nodes, respectively.
- **SelfGAT:** 3 graph self-attention layers with 16, 32 and 64 features, respectively, ending with a dropout layer with a dropout probability of 20%. The dropout layer's output is aggregated using a global addition pooling layer and fed to an MLP. This MLP is formed by 3 fully connected layers with 192, 96 and N nodes, respectively, with a 20% dropout probability layers between each pair.
- **Transformer:** 3 graph transformer layers with 16, 32 and 64 features, respectively, ending with a dropout layer with a dropout probability of 20%. The dropout layer's output is aggregated using a global addition pooling layer and fed to an MLP. This MLP is formed by 3 fully connected layers with $64H$, $32H$ and N nodes, respectively, with H being the number of heads, and with a 20% dropout probability layers between each pair. A version with a fourth transformer layer with 32 features is also evaluated. Besides, other variants are also explored, namely:
 - **Transformer (more features):** transformer architecture with the first layer having 32 features instead of 16.
 - **Transformer (no dropout):** transformer architecture without the dropout layers.
 - **Transformer + GCN:** transformer architecture adding a graph convolutional layer after the three transformer layers.

4.4. Validation

In order to evaluate the classification performance of the proposed GNN architectures, we used a cross-validation method. In particular, we applied the leave-one-out cross-validation (LOOCV) to investigate the proposed system. LOOCV is a special case of k -fold cross-validation where the number of folds is equal to the number of patients in the dataset. Each patient's data is held out as the test set while the

remaining patients' data are used as the training set. This process is repeated for each patient, resulting in a comprehensive evaluation across all patients. The main purpose of employing LOOCV is to assess the generalization capability of the model by evaluating its performance on unseen patient data. Additionally, LOOCV offers a patient-centric evaluation, allowing us to assess the model's performance for each individual in the dataset.

4.5. Explainability

A significant advantage of GNNs compared to other machine learning approaches to EEG analysis is the high capacity of explainability they possess: since they use graphs to process their data and graph structures can be easily mapped to concepts comprehensible by humans (in this case, an EEG), it is relatively simple to determine a model's *reasoning* behind a prediction. To this end, PyTorch Geometric includes a functionality which allows the use of the GNNExplainer algorithm ([Ying et al., 2019](#)). The algorithm is able to deliver an analysis on feature, nodes and edge importance, assigning an importance score to every selected element. In this case, the algorithm can be useful to determine which nodes and which attributes had a larger relevance for a prediction and how important the relationship (edge) between two nodes was more significant during the calculations. This is used in this paper for both AD diagnosis and seizure discrimination use cases.

5. Results

Next, we delve into the analysis of the results obtained for the two use cases, namely, AD diagnosis (Section 5.1) and discrimination of epileptic seizures (Section 5.2). We show the evaluation of the different GNN alternatives, the cross-validation performed on the best GNN architecture, the comparison with the state of the art, and the results of the feature/node/edge importance for explainability.

5.1. Results for detection of Alzheimer's Disease

Design Space Exploration. As starting point of the analysis, we explored the design space in terms of GNN model architecture (see [Table 2](#)) and feature extraction mechanisms. In this case, only the binary classification task (Alzheimer's against healthy control) is evaluated, without cross-validation. [Table 7](#) at the appendix lists the results for all the evaluated architectures, whereas [Fig. 4](#) provides a summary by showing the test accuracy against the number of trainable parameters for the different explored GNN architectures. The results show the superiority of techniques with trainable edge weights over the vanilla GCN. The best performance is obtained with a transformer of three layers and three heads; adding more heads, layers, or features does not provide a noticeable advantage. Overall, this architecture achieves an advantage of over 5% with respect to the best GCN, which uses PSD as node features and CSD as edge feature. However, the superior performance of transformers comes at the expense of a higher number of trainable weights, around a 10× more than GCN.

For a comparison with neural networks not based on graphs, an MLP was trained as a baseline. We considered 4 fully-connected linear layers with a ReLU activation function, a batch size of 128, cross-entropy loss, an Adam optimizer and the PSD of all nodes as input features. The learning rate and the hidden layer size being selected as hyperparameters using grid search; the best values were 0.001 and 256, respectively, yielding a test accuracy of 61.05% with over 161K parameters. For the comparison with a state-of-the-art model, we include the CNN from [Ieracitano et al. \(2019\)](#), which leads to a test accuracy around 90% but a hefty model with over 3 million parameters. Compared to these points, transformers achieve a very good balance between accuracy and computational efficiency.

Cross-validation. In light of the results of the design space exploration, we took the GNN architecture leading to the highest accuracy

Table 3

Results for the binary and ternary classifications of Alzheimer's disease with cross-validation for the best performing GNN architecture.

	Set	Average accuracy	Minimum accuracy	Maximum accuracy	Standard deviation
Binary	Train	85.18%	64.15%	92.23%	7.95%
	Test	91.77%	83.10%	100%	4.68%
Ternary	Train	88.10%	55.13%	99.15%	9.64%
	Test	89.02%	60.94%	100%	9.85%

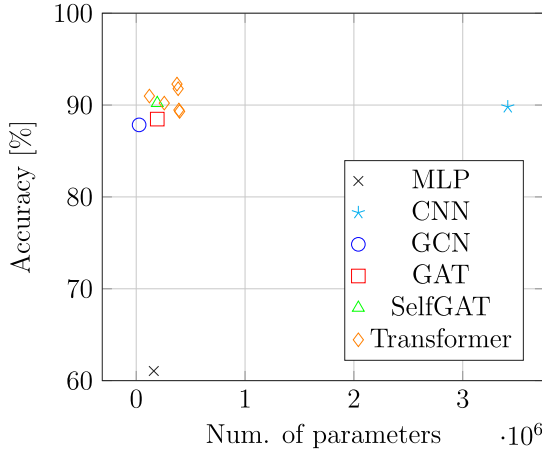


Fig. 4. Number of trainable parameters and accuracy of the proposed GNN architectures in the Alzheimer's use case.

(transformer, 3 layers, 3 heads) and performed the cross-validation both for the binary (AD vs HC) and ternary classifications (AD vs MCI vs HC). [Table 3](#) shows a summary of the statistics of the cross-validation for both the train and test splits in both the binary and ternary classification tasks.

Compared to the results obtained with the original validation method, the accuracy for both the binary and the ternary cases is noticeably lower, showing that the performance of the model for unseen patients is lower, which is to be expected. However, both cases show that the proposed model can still generally predict the diagnosis of EEG data belonging to a new patient properly, displaying its high generalization power. Interestingly, our proposal works especially well for the ternary classification task, implying that the chosen model is particularly good at discriminating healthy patients from those suffering from AD or MCI, since the inclusion of data belonging to the HC class appears to improve the general performance.

Comparison with the State of the Art. [Table 4](#) shows the comparison between our results and those of related work, where the best results for both the binary and ternary classification cases are achieved by [Ieracitano et al. \(2020\)](#) with MLPs, yet with very complex feature extraction of multiple statistics from continuous wavelet transform and bispectrum representations. In this landscape, it is observed how our GNN is able to directly compete with most alternatives except for [Ieracitano et al. \(2020\)](#), which prove to be slightly superior. For binary classification, in particular, our graph-based transformer design achieves a remarkable 91.77% test accuracy, which is close to the best accuracy, thanks to its ability to learn the correlations between nodes. For ternary classification, our graph-based transformer can compete with the state of the art, achieving an accuracy of 89.02% very close to that of [Ieracitano et al. \(2020\)](#) yet with a much simpler feature extraction and including increased explainability, as we will see later in the paper. Besides, it is also worth pointing out that the accuracy for both binary and ternary classification is obtained using a patient-centric LOOCV, which may lead to a lower yet more realistic score than studies using other methods, such as the k -fold (70/30) patient-agnostic validation of [Ieracitano et al. \(2020\)](#), hence putting our model at disadvantage.

Table 4

Comparison of machine learning methods for the diagnosis of Alzheimer's disease using EEGs.

Reference	Method	Classification	Accuracy
Ieracitano et al. (2020)	MLP	Binary	96.95%
Ieracitano et al. (2020)	SVM	Binary	91.88%
Ieracitano et al. (2019)	CNN	Binary	89.8%
Ieracitano et al. (2019)	KNN	Binary	92.95%
Morabito et al. (2016)	Random Forest	Binary	85%
Morabito et al. (2016)	Decision Tree	Binary	82%
Zhao and He (2015)	SVM	Binary	92%
Fan et al. (2018)	LR	Binary	82%
Morabito et al. (2016)	CNN	Ternary	82%
Ieracitano et al. (2019)	CNN	Ternary	83.3%
Ieracitano et al. (2020)	MLP	Ternary	89.22%
Klepl et al. (2022)	GNN	Binary	92%
Our work	GNN	Binary	91.77%
Our work	GNN	Ternary	89.02%

Feature Importance. To facilitate the explainability analysis, one of the subjects with AD from the dataset was left outside training, and their data was used for calculating feature, node and edge importance for the model's predictions, which are shown in [Fig. 5](#). Firstly, the feature importance is analyzed. We can observe in the left plot how the features corresponding to the alpha band are the most relevant with significant difference. In terms of location, we can see in the center plot how the most relevant nodes are those associated with electrodes placed in the occipital (O1) and pre-frontal (Fp1) areas. In the right plot, we observe how the most important edges are all originating from the O1 electrode, suggesting that the importance lies more on the nodes. In any case, both node and edge importance plots show a relatively narrow range of importance. For a discussion on the clinical relevance of these results, we refer the reader to [Section 6](#).

5.2. Results for discrimination of epilepsy seizures

Design Space Exploration. As shown above in [Section 5.1](#), we first explore the design space in terms of GNN model architecture and feature extraction mechanisms for binary classification without cross-validation. [Table 8](#) in the appendix lists the results for all the evaluated architectures, whereas [Fig. 6](#) provides a summary by showing the test accuracy against the number of trainable parameters of each model. The results are similar to the AD use case: the best GCN uses PSD and CSD as node and edge features, respectively, but they are still around 5% worse in terms of accuracy than the best transformer models. SelfGAT and the variants of transformers without dropout and more features achieve the highest performance, over 92% of accuracy. Interestingly, the transformer model with 3 layers and 3 heads achieves a very competitive performance in this use case as well. In all cases, the increase in performance comes with a much higher number of trainable parameters, with the exception of SelfGAT, but all GNN architectures vastly outperform the MLP baseline, showing the advantage of leveraging GNNs for this task.

Like in the case of AD, we compare the performance of our models with an MLP baseline and a competitive model from the literature.

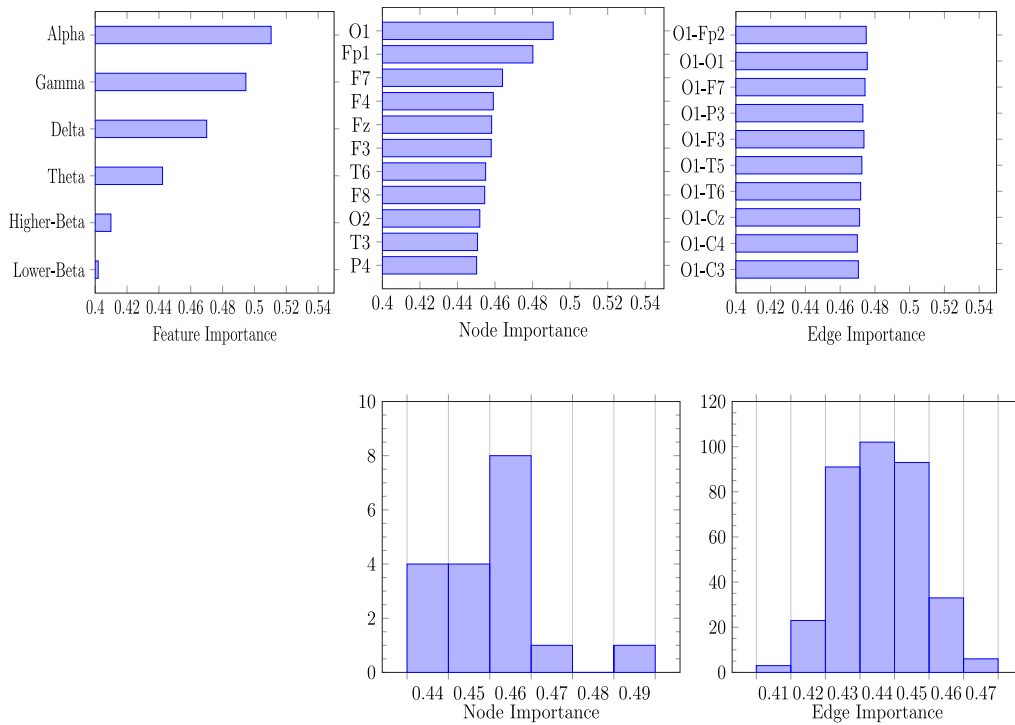


Fig. 5. Average importance for features (frequency bands), nodes, and edges for one subject's predictions in the Alzheimer's disease use case. Bottom plots show the histogram of the node and edge importance values.

Table 5

Results for the binary and ternary classifications of the seizure discrimination use case with cross-validation for the best performing GNN architecture.

	Set	Average accuracy	Minimum accuracy	Maximum accuracy	Standard deviation
Binary	Train	99.73%	95.31%	99.98%	0.98%
	Test	99.84%	93.85%	100%	0.71%
Ternary	Train	96.69%	91.88%	98.76%	1.70%
	Test	96.95%	90.86%	99.59%	1.66%

Again, the MLP consists of 4 fully-connected layers, ReLU activation function, a batch size of 128, cross-entropy loss, an Adam optimizer and the PSD of all nodes as input features. The learning rate is 0.001 and the hidden layer size is 256, which yields a test accuracy of 56% with over 150K parameters. For the competitive baseline, we include the CNN model from Giudice, Varone, et al. (2022) with a test accuracy over 90% but more than 600K parameters. Again, compared to these points, transformers achieve a very good balance between accuracy and computational efficiency.

Cross-Validation. An additional experiment was performed taking the best transformer model and cross-validating the results with LOOCV considering both the binary and ternary classification tasks. The results, summarized on Table 5, show an excellent test accuracy of 99.84% for binary classification and 96.95% for ternary classification, with a relatively small standard deviation across patients.

Comparison with the State of the Art. Next, we present a comparative analysis of our proposed model against existing state-of-the-art techniques for EEG-based seizure classification. Table 6 summarizes the results obtained by various methods with various datasets. Notably, our model stands out as one of the best-performing alternatives for binary classification with an accuracy beyond 99%, outperforming the best alternative from Giudice, Varone, et al. (2022) that achieves a 94.4% using CNNs. For ternary classification, which includes healthy patients as a separate category, our model continues to shine with an accuracy of 96.95%, which is over 10 points above the closest contender in this

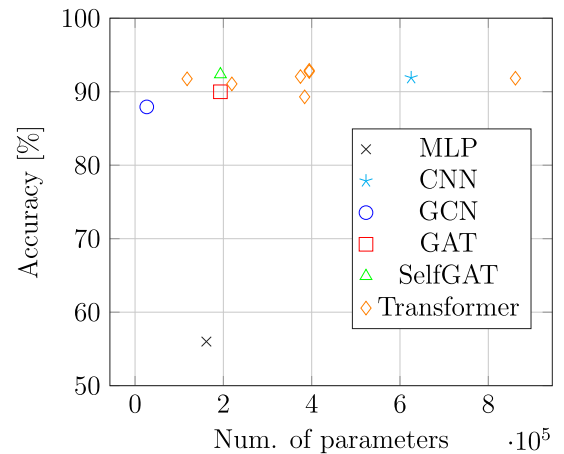


Fig. 6. Number of trainable parameters and accuracy of the proposed GNN architectures in the epilepsy use case.

category (Giudice, Ferlazzo, et al., 2022), which attains an accuracy of 85.7%. It is worth noting the comparison with Giudice, Ferlazzo, et al. (2022) is fair as we use the same dataset to train and test. Therefore, these results underscore the effectiveness and robustness of our model, especially in scenarios requiring comprehensive seizure classification.

Table 6

Comparison of machine learning methods for the discrimination of epileptic seizures using EEGs.

Reference	Method	Classification	Accuracy
Giudice, Varone, et al. (2022)	CNN	Binary	94.4%
Giudice, Ferlazzo, et al. (2022)	CNN	Ternary	85.7%
Ahmadi et al. (2020)	KNN	Binary	82.2%
Ahmadi et al. (2020)	SVM	Binary	75.8%
Ahmadi et al. (2020)	Random Forest	Binary	77.6%
Ahmadi et al. (2020)	Decision Tree	Binary	73.8%
Ahmadi et al. (2020)	GBoost	Binary	78.2%
Zsom et al. (2019)	XGBoost	Binary	78%
Our work	GNN	Binary	99.84%
Our work	GNN	Ternary	96.95%

Feature Importance As in previous sections, to aid in the explainability analysis, one of the subjects with epilepsy from the dataset was left outside training, and their data was used for calculating node, feature and edge importance for the model's predictions. The results are summarized in Fig. 7. As seen in the left plot showing feature importance, the Theta band appears to be the most important band for these predictions, with a noticeable difference with respect to other bands, especially Beta and Delta. Further, as shown in the center plot, the most relevant nodes are those associated with electrodes located in the back of the patient's head during an EEG recording, such as T6, O1 and O2. Therefore, the data from these areas of the brain appears to be relevant when classifying seizure types. However, these relevance differences do not appear to be significant, as the observed standard deviation of these is just 0.00435 (see the histogram in the bottom center plot of Fig. 7), so there is no conclusive evidence of these nodes being important for all predictions. Finally, the relevance of the edges was also calculated as seen on the right plot of Fig. 7, where the node relationships with the highest scores are those adjacent to the node associated to electrode O2, located at the subject's backhead. This aligns with the discoveries made for node importance. In addition, these scores show a standard deviation of 0.0114, which is significantly more relevant than that of the node importance, so these results seem to show us a more reliable explanation for this subject's predictions by the proposed GNN model. For a discussion on the clinical relevance of these findings, we refer the reader to the next section.

6. Discussion

Accuracy. The current study uses GNNs to solve two difficult challenges: classifying EEG recordings of patients with AD or MCI, as well as patients with ES, PNES, and differentiating them from healthy control patients. The results of our experiments highlight the superiority of GNNs in classifying various EEG challenges compared to traditional methods. Our proposed GNN delivers greater accuracy performance due to its capacity to capture the intricate spatial and temporal relationships between the human brain, which is effectively a complex network of interconnected neurons whose activity can be recorded by electrodes structured as graph nodes. Graphs excel at representing this interconnection, allowing for the investigation of linkages, routes, and groupings. While CNNs have been widely applied to EEG classification, their use implies that EEG relational data is compressed into a pseudo-image. Given that different brain regions are not necessarily associated by scalp-distance (thus close electrodes), CNNs do not seem to be the best option. CNNs therefore fall short in capturing the extremely parallel and interconnected nature of brain function. Conversely, GNNs find connections between far-off electrodes via intricate pathways that are not accessible to traditional convolutional models. So the dynamic nature of brain networks, reflected in the structure of evolving graphs,

has immense potential to unravel the mechanisms underlying these diseases.

Versatility. Another key aspect of our work is the choice of a single model for classifying two different pathologies. This approach simplifies the fragmented landscape of AI applied to neurological diseases. Our transformer-based GNN proves to be versatile, adapting to the study of different phenomena that contribute to the definition of a specific disease. For instance, Alzheimer's disease manifests with altered graph patterns, reflecting the progressive disruption of neural connectivity. These changes, which occur on relatively long time scales, can provide crucial clues for disease classification. In the case of epilepsy, the study captures the transformation of information flow from a localized source to a diffuse seizure. This flood of information, captured by dynamic changes in the graph, albeit in a shorter scale than for Alzheimer's disease, could serve as a valuable biomarker for seizure prediction and classification. This multidisciplinary approach highlights the advantages of combining EEG-based graph analysis with medical expertise. By integrating these perspectives, researchers can gain a deeper understanding of the underlying pathophysiology of neurological disorders and develop more effective diagnostic and therapeutic strategies.

Explainability. In addition to surpassing previous methods in terms of accuracy, our approach has allowed us to extract feature importance information that is distinct and unique to each disease. On the one hand, in the analysis of Alzheimer's (Fig. 5), the alpha band emerges as the most discriminating, in line with scientific evidence. In particular, studies have shown that the alpha band is the dominant rhythm in healthy adults during a relaxed waking state, and this is consistent with our results (Babiloni et al., 2009; Brueggen et al., 2017). On the other hand, in the study of epilepsy (Fig. 7), the distributed nature of epileptic phenomena due to neuronal synchronization would suggest that no feature would stand out. However, on the PNES side, some studies have observed anomalies in the Gamma band (Umesh, Tikka, Goyal, Sinha, & Nizamie, 2017) and also a decrease of amplitude in the Theta band, for measurements done in the posterior parietal cortex (Arzy et al., 2014). This is moderately aligned with our results of feature importance and also on node importance (T6, O1, O2), although the latter is unclear due to the low variance on node importance values.

Generalizability. In spite of the arguments made with respect to the accuracy, versatility, and explainability advantages of GNNs, a question still arises with respect to the potential generalizability of the approach. The current study considers a wide demographic, and judging by the cross-validation results, the model generalizes well across patients thanks to its capacity to extract patterns between arbitrary nodes – without the need for explicitly using information such as age or other indicators. However, should the use case require it, GNNs can be augmented with global features either as global nodes in the dataset graph, or extra features in the final classification layers. Beyond this, GNNs have shown the capacity to scale to billions of nodes and samples without changing its underlying structure (Wu et al., 2022). Hence, we posit that the proposed methods could be used in larger graphs (e.g. for 64-channel EEGs) and larger datasets. Regarding the latter, our recent study (Sánchez Bonilla, 2024) has achieved similar accuracy than the present work in the seizure discrimination use case using the TUEP dataset from the TUH corpus (Harati et al., 2014), which gathers double the number of patients than the present study.

Real-time Operation. Both use cases presented in this study assume data acquisition in a resting state and do not have any real-time processing and inference requirements. However, it is worth noting that the proposed pipeline would not present any foreseeable hardware constraints for real-time operation, should automation and integration into existing hospital systems be required. In particular, the pre-processing uses standard filtering and transformations that can be easily computed with off-the-shelf hardware or implemented in custom hardware,

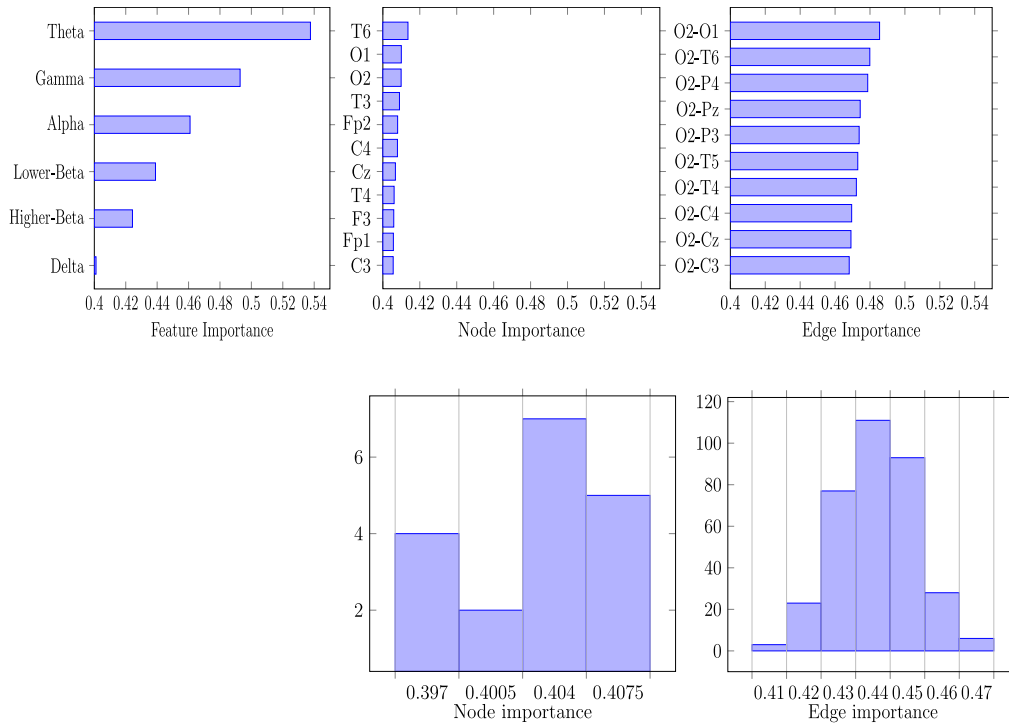


Fig. 7. Average importance for features (frequency bands), nodes, and edges for one subject's predictions in the epilepsy use case. Bottom plots show the histogram of the node and edge importance values.

if needed. The GNN model is lightweight given the low number of nodes (19), edges (19×18) and features per node (6). At this scale, GNN inference is not a bottleneck in off-the-shelf hardware, but could be even reduced further with existing GNN hardware accelerators, if necessary (Abadal, Jain, Guirado, López-Alonso, & Alarcón, 2021).

7. Conclusions

This paper is built on the hypothesis that GNNs have a higher expressive power over existing techniques such as CNNs or LSTMs for the interpretation of EEG signals. In this study, we have empirically proven the hypothesis for two use cases, i.e. Alzheimer's disease and epilepsy classification. After a design space exploration, we found that graph transformers can achieve very high performance for both use cases. The same model matches the performance of the state of the art in Alzheimer's disease diagnosis and outperforms the best model in epilepsy seizure discrimination by several percentage points, without the need for very intricate input features or complex models, hence remaining very computationally efficient. In fact, graph transformers learn the correlations between nodes, hence completely eliminating the need to define and calculate edge weights in advance. Finally, we also posit that GNNs provide a higher explainability potential, and show that the results obtained with the graph transformers seem to match with the expert knowledge for both use cases. Therefore, our work represents a significant step forward in the analysis of EEG signals using artificial intelligence and GNNs are confirmed as a valuable tool for classifying various neurological pathologies, opening up new prospects for research and clinical practice.

CRedit authorship contribution statement

Sergi Abadal: Project administration, Writing – original draft, Writing – review & editing. **Pablo Galván:** Investigation, Validation. **Alberto Mármol:** Investigation, Validation. **Nadia Mammone:** Conceptualization, Validation. **Cosimo Ieracitano:** Methodology, Writing –

review & editing. **Michele Lo Giudice:** Methodology, Writing – original draft, Writing – review & editing. **Alessandro Salvini:** Project administration, Supervision. **Francesco Carlo Morabito:** Conceptualization, Data curation, Project administration, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Performance tables

To avoid clutter in the main document, we hereby include the tables listing all the results obtained in the design space exploration of GNN models and feature extraction. Table 7 presents the results for the Alzheimer's disease classification, whereas Table 8 details the results for the seizure discrimination problem. In both cases, the results correspond to the binary task without cross-validation, and include performance indicators such as accuracy (Acc.), balanced accuracy (Bal.), precision (Prec.), recall, F1 score, and area under the ROC curve (AUC).

Data availability

The data that has been used is confidential.

Table 7Comparison of results for the binary classification in Alzheimer's diseases detection without cross-validation. *Bal.* is balanced accuracy. *Prec.* is precision.

Model	Heads	Layers	Nodes	Edges	Set	Acc.	Bal.	Prec.	Recall	F1	AUC
GCN	–	3	PSD	CSD	Train	91.37	88.91	90.34	88.91	89.57	0.96
					Val	88.48	84.62	87.04	84.62	85.69	0.93
					Test	87.84	84.22	85.57	84.22	84.85	0.93
	–	3	PSD	PCC	Train	86.32	85.78	84.91	85.78	84.66	0.94
					Val	79.47	79.83	79.94	79.83	80.01	0.89
					Test	78.22	78.09	77.46	78.09	77.57	0.87
	–	3	PSD	PLI	Train	93.26	87.48	91.23	87.48	88.93	0.91
					Val	83.50	76.65	81.17	76.65	78.35	0.88
					Test	82.06	75.09	79.68	75.09	76.74	0.87
	–	3	Stats	CSD	Train	90.97	91.44	88.51	91.44	89.70	0.97
					Val	81.52	81.31	81.29	81.31	81.50	0.94
					Test	80.07	80.89	80.96	80.89	80.15	0.95
GAT	3	3	PSD	Att	Train	92.74	89.27	91.11	89.27	90.14	0.97
					Val	85.83	80.30	82.43	80.30	81.26	0.92
					Test	88.47	83.33	84.17	83.33	83.74	0.93
SelfGAT	3	3	PSD	Att	Train	98.24	97.67	98.12	97.67	97.89	0.99
					Val	91.14	89.01	89.69	89.01	89.34	0.94
					Test	90.21	87.94	88.67	87.94	88.29	0.94
Transformer	2	3	PSD	Att	Train	96.02	96.39	95.90	96.39	96.14	0.99
					Val	92.89	92.58	91.41	92.58	91.97	0.97
					Test	90.98	91.39	91.08	91.39	91.23	0.96
	3	3	PSD	Att	Train	99.20	99.10	99.06	99.10	99.08	0.99
					Val	93.11	91.60	91.70	91.60	91.63	0.97
					Test	92.28	90.81	90.47	90.81	90.64	0.97
	4	3	PSD	Att	Train	99.68	99.64	99.52	99.64	99.58	0.99
					Val	92.26	91.79	91.96	91.79	91.84	0.97
					Test	91.78	90.51	90.34	90.51	90.42	0.97
	3	4	PSD	Att	Train	98.83	98.83	98.82	98.83	98.82	1
					Val	90.27	90.27	90.24	90.27	90.25	0.96
					Test	89.30	89.28	89.23	89.28	89.25	0.96
Transformer (more feat.)	3	3	PSD	Att	Train	98.24	97.67	98.12	97.67	97.89	0.99
					Val	91.14	89.01	89.69	89.01	89.34	0.94
					Test	90.21	87.94	88.67	87.94	88.29	0.94
Transformer (no dropout)	4	3	PSD	Att	Train	95.70	94.74	94.90	94.74	94.82	0.99
					Val	89.75	87.46	88.32	87.46	87.87	0.95
					Test	89.46	87.27	87.98	87.27	87.61	0.94
Transformer + GCN	3	4	PSD	Att	Train	99.26	99.57	99.78	99.57	99.66	1
					Val	90.25	90.07	89.97	90.07	90.32	0.97
					Test	90.74	90.89	91.07	90.89	90.90	0.97

References

- Abadal, S., Jain, A., Guirado, R., López-Alonso, J., & Alarcón, E. (2021). Computing graph neural networks: A survey from algorithms to accelerators. *ACM Computing Surveys*, 54(9), 1–38.
- Aboalayon, K. A. I., Faezipour, M., Almuhammadi, W. S., & Moslehpour, S. (2016). Sleep stage classification using EEG signal analysis: A comprehensive survey and new investigation. *Entropy*, 18(9), 272.
- Aggarwal, S., & Chugh, N. (2022). Review of machine learning techniques for EEG based brain computer interface. *Archives of Computational Methods in Engineering*, 1–20.
- Ahmadi, N., Pei, Y., Carrette, E., Aldenkamp, A. P., & Pechenizkiy, M. (2020). EEG-based classification of epilepsy and PNES: EEG microstate and functional brain network features. *Brain Informatics*, 7(1), 6. <http://dx.doi.org/10.1186/s40708-020-00107-z>.
- American Psychiatric Association, D., Association, A. P., et al. (2013). *Diagnostic and statistical manual of mental disorders: DSM-5: vol. 5*, (no. 5), American Psychiatric Association Washington, DC.
- Arzy, S., Halje, P., Schechter, D. S., Spinelli, L., Seeck, M., & Blanke, O. (2014). Neural generators of psychogenic seizures: Evidence from intracranial and extracranial brain recordings. *Epilepsy & Behavior*, 31, 381–385.
- Babiloni, C., Frisoni, G. B., Pievani, M., Vecchio, F., Lizio, R., Buttiglione, M., et al. (2009). Hippocampal volume and cortical sources of EEG alpha rhythms in mild cognitive impairment and Alzheimer disease. *NeuroImage*, 44(1), 123–135. <http://dx.doi.org/10.1016/j.neuroimage.2008.08.005>.
- Bai, L., Cui, L., Wang, Y., Li, M., Li, J., Philip, S. Y., et al. (2024). HAQJSK: Hierarchical-aligned quantum Jensen-Shannon kernels for graph classification. *IEEE Transactions on Knowledge and Data Engineering*.
- Bi, X., & Wang, H. (2019). Early Alzheimer's disease diagnosis based on EEG spectral images using deep learning. *Neural Networks*, 114, 119–135. <http://dx.doi.org/10.1016/j.neunet.2019.02.005>.
- Brueggen, K., Fiala, C., Berger, C., Ochmann, S., Babiloni, C., & Teipel, S. J. (2017). Early changes in alpha band power and DMN BOLD activity in Alzheimer's disease: A simultaneous resting state EEG-fMRI study. *Frontiers in Aging Neuroscience*, 9, 319.
- Brunner, C., Delorme, A., & Makeig, S. (2013). Eeglab—an open source Matlab toolbox for electrophysiological research. *Biomedical Engineering/Biomedizinische Technik*, 58(S1-1-Track-G), Article 000010151520134182.
- Cassani, R., Estarellas, M., San-Martin, R., Fraga, F. J., & Falk, T. H. (2018). Systematic review on resting-state EEG for Alzheimer's disease diagnosis and progression assessment. *Disease Markers*, 2018.

Table 8Comparison of results for the binary classification of seizure discrimination, without cross-validation. *Bal.* is balanced accuracy. *Prec.* is precision.

Model	Heads	Layers	Nodes	Edges	Set	Acc.	Bal.	Prec.	Recall	F1	AUC
GCN	–	3	PSD	CSD	Train	97.03	97.03	97.02	97.03	97.02	1
					Val	88.38	88.31	88.34	88.31	88.32	0.95
					Test	87.93	87.84	87.89	87.84	87.86	0.96
	–	3	PSD	PCC	Train	84.10	84.09	84.02	84.09	84.05	0.93
					Val	79.64	79.59	79.64	79.59	79.61	0.88
					Test	77.77	77.71	77.72	77.71	77.71	0.86
	–	3	PSD	PLI	Train	90.06	90.05	90.02	90.05	90.03	0.97
					Val	80.20	80.15	80.15	80.15	80.15	0.88
					Test	81.74	81.56	81.56	81.56	81.56	0.89
	–	3	Raw	CSD	Train	62.97	62.3	62.98	62.3	62.13	0.69
					Val	55.87	55.34	55.77	55.34	54.75	0.58
					Test	55.84	54.71	54.93	54.71	54.57	0.58
	–	3	Stats	CSD	Train	84.98	84.83	85.01	84.83	84.9	0.93
					Val	82.62	82.49	82.6	82.49	82.53	0.91
					Test	81.89	81.58	81.86	81.58	81.69	0.9
GAT	3	3	PSD	Att	Train	99.90	99.90	99.90	99.90	99.90	1
					Val	90.24	90.24	90.15	90.24	90.19	0.96
					Test	89.99	89.98	89.96	89.98	89.97	0.97
SelfGAT	3	3	PSD	Att	Train	100.00	100.00	100.00	100.00	100.00	1
					Val	92.56	92.55	92.51	92.55	92.53	0.97
					Test	92.36	92.31	92.34	92.31	92.33	0.97
Transformer	2	3	PSD	Att	Train	99.99	99.99	99.99	99.99	99.99	1
					Val	92.48	92.50	92.40	92.50	92.44	0.97
					Test	91.75	91.77	91.70	91.77	91.73	0.98
	3	3	PSD	Att	Train	99.99	99.99	99.99	99.99	99.99	1
					Val	92.79	92.79	92.76	92.79	92.78	0.98
					Test	92.06	91.96	91.99	91.96	91.97	0.98
	4	3	PSD	Att	Train	100.00	100.00	100.00	100.00	100.00	1
					Val	91.85	91.74	91.86	91.74	91.79	0.97
					Test	91.83	91.80	91.82	91.80	91.81	0.98
	3	4	PSD	Att	Train	98.83	98.83	98.82	98.83	98.82	1
					Val	90.27	90.27	90.24	90.27	90.25	0.96
					Test	89.30	89.28	89.23	89.28	89.25	0.96
Transformer (more feat.)	3	3	PSD	Att	Train	99.99	99.99	99.99	99.99	99.99	1
					Val	92.69	92.62	92.70	92.62	92.66	0.98
					Test	92.74	92.69	92.72	92.69	92.70	0.98
Transformer (no dropout)	3	3	PSD	Att	Train	100.00	100.00	100.00	100.00	100.00	1
					Val	93.07	93.03	93.04	93.03	93.04	0.98
					Test	92.90	92.93	92.89	92.93	92.89	0.98
Transformer +GCN	3	4	PSD	Att	Train	99.69	99.69	99.68	99.69	99.69	1
					Val	91.62	91.61	91.58	91.61	91.60	0.97
					Test	91.06	91.10	91.01	91.10	91.04	0.97

Chaturvedi, M., Bogaarts, J. G., Kozak, V. V., Hatz, F., Gschwandtner, U., Meyer, A., et al. (2019). Phase lag index and spectral power as QEEG features for identification of patients with mild cognitive impairment in Parkinson's disease. *Clinical Neurophysiology*, 130(10), 1937–1944.

Chikhi, S., Matton, N., & Blanchet, S. (2022). EEG power spectral measures of cognitive workload: A meta-analysis. *Psychophysiology*, 59(6), Article e14009.

Curley, W. H., Forgacs, P. B., Voss, H. U., Conte, M. M., & Schiff, N. D. (2018). Characterization of EEG signals revealing covert cognition in the injured brain. *Brain*, 141(5), 1404–1421.

Da Silva, F. L. (2023). EEG: Origin and measurement. In *EEG-FMRI: physiological basis, technique, and applications* (pp. 23–48). Springer.

Faiman, I., Smith, S., Hodson, J., Young, A. H., & Shotbolt, P. (2021). Resting-state EEG for the diagnosis of idiopathic epilepsy and psychogenic nonepileptic seizures: A systematic review. *Epilepsy & Behavior*, 121, Article 108047.

Fan, M., Yang, A. C., Fuh, J.-L., & Chou, C.-A. (2018). Topological pattern recognition of severe Alzheimer's disease via regularised supervised learning of EEG complexity. *Frontiers in Neuroscience*, 12, 1–10. <http://dx.doi.org/10.3389/fnins.2018.00685>.

Fey, M., & Lenssen, J. E. (2019). Fast graph representation learning with PyTorch geometric. arXiv preprint [arXiv:1903.02428](https://arxiv.org/abs/1903.02428).

Fouladi, S., Safaei, A. A., Mammone, N., Ghaderi, F., & Ebadi, M. (2022). Efficient deep neural networks for classification of Alzheimer's disease and mild cognitive impairment from scalp EEG recordings. *Cognitive Computation*, 14(4), 1247–1268.

Gu, X., Cao, Z., Jolfaei, A., Xu, P., Wu, D., Jung, T.-P., et al. (2021). EEG-based brain-computer interfaces (BCIs): A survey of recent studies on signal sensing technologies and computational intelligence approaches and their applications. *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, 18(5), 1645–1666.

Harati, A., Lopez, S., Obeid, I., Picone, J., Jacobson, M., & Tobochnik, S. (2014). The TUH EEG CORPUS: A big data resource for automated EEG interpretation. In *2014 IEEE signal processing in medicine and biology symposium*. IEEE.

Heyat, M. B. B. (2017). *Insomnia: Medical sleep disorder & diagnosis*. Anchor Academic Publishing.

Hou, Y., Jia, S., Lun, X., Hao, Z., Shi, Y., Li, Y., et al. (2022). GCNs-net: A graph convolutional neural network approach for decoding time-resolved eeg motor imagery signals. *IEEE Transactions on Neural Networks and Learning Systems*.

Ieracitano, C., Mammone, N., Bramanti, A., Hussain, A., & Morabito, F. C. (2019). A convolutional neural network approach for classification of dementia stages based on 2D-spectral representation of EEG recordings. *Neurocomputing*, 323, 96–107. <http://dx.doi.org/10.1016/j.neucom.2018.09.071>.

Ieracitano, C., Mammone, N., Hussain, A., & Morabito, F. C. (2020). A novel multi-modal machine learning based approach for automatic classification of EEG recordings in dementia. *Neural Networks*, 123, 176–190. <http://dx.doi.org/10.1016/j.neunet.2019.12.006>.

Jasper, H. H. (1983). The ten-twenty electrode system of the international federation. Recommendations for the Practice of Clinical Neurophysiology, Elsevier Publishing Company.

Khademi, Z., Ebrahimi, F., & Kordy, H. M. (2022). A transfer learning-based CNN and LSTM hybrid deep learning model to classify motor imagery EEG signals. *Computers in Biology and Medicine*, 143, Article 105288.

Kim, D., & Kim, K. (2018). Detection of early stage Alzheimer's disease using EEG relative power with deep neural network. In *IEEE engineering in medicine and biology society annual int. conf.*

Kim, D., & Oh, A. (2021). How to find your friendly neighborhood: Graph attention design with self-supervision. In *International conference on learning representations*.

- Kipf, T. N., & Welling, M. (2016). Semi-supervised classification with graph convolutional networks. CoRR abs/1609.02907, arXiv:1609.02907, URL <http://arxiv.org/abs/1609.02907>.
- Klepl, D., He, F., Wu, M., Blackburn, D. J., & Sarrigiannis, P. (2022). Eeg-based graph neural network classification of Alzheimer's disease: An empirical evaluation of functional connectivity methods. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 30, 2651–2660.
- Kuanar, S., Athitsos, V., Pradhan, N., Mishra, A., & Rao, K. R. (2018). Cognitive analysis of working memory load from EEG, by a deep recurrent neural network. In *2018 IEEE international conference on acoustics, speech and signal processing* (pp. 2576–2580). IEEE.
- Lehmann, C., Koenig, T., Jelic, V., Prichep, L., John, R. E., Wahlund, L.-O., et al. (2007). Application and comparison of classification algorithms for recognition of Alzheimer's disease in electrical brain activity (EEG). *Journal of Neuroscience Methods*, 161(2), 342–350.
- Li, M., Micheli, A., Wang, Y. G., Pan, S., Lió, P., Gnecco, G. S., et al. (2024). Guest editorial: Deep neural networks for graphs: Theory, models, algorithms, and applications. *IEEE Transactions on Neural Networks and Learning Systems*, 35(4), 4367–4372.
- Li, M., Zhou, S., Chen, Y., Huang, C., & Jiang, Y. (2024). EduCross: Dual adversarial bipartite hypergraph learning for cross-modal retrieval in multimodal educational slides. *Information Fusion*, 109, Article 102428.
- Lo Giudice, M., Ferlazzo, E., Mammone, N., Gasparini, S., Cianci, V., Pascarella, A., et al. (2022). Convolutional neural network classification of rest EEG signals among people with epilepsy, psychogenic non epileptic seizures and control subjects. *International Journal of Environmental Research and Public Health*, 19(23), <http://dx.doi.org/10.3390/ijerph192315733>.
- Lo Giudice, M., Varone, G., Ieracitano, C., Mammone, N., Tripodi, G. G., Ferlazzo, E., et al. (2022). Permutation entropy-based interpretability of convolutional neural network models for interictal eeg discrimination of subjects with epileptic seizures vs. psychogenic non-epileptic seizures. *Entropy*, 24(1), 102.
- McBride, J. C., Zhao, X., Munro, N. B., Smith, C. D., Jicha, G. A., Hively, L., et al. (2014). Spectral and complexity analysis of scalp EEG characteristics for mild cognitive impairment and early Alzheimer's disease. *Computer Methods and Programs in Biomedicine*, 114(2), 153–163.
- Min, E., Chen, R., Bian, Y., Xu, T., Zhao, K., Huang, W., et al. (2022). Transformer for graphs: An overview from architecture perspective. arXiv preprint arXiv:2202.08455.
- Morabito, F. C., Campolo, M., Ieracitano, C., Ebadi, J. M., Bonanno, L., Bramanti, A., et al. (2016). Deep convolutional neural networks for classification of mild cognitive impaired and Alzheimer's disease patients from scalp EEG recordings. In *2016 IEEE 2nd international forum on research and technologies for society and industry leveraging a better tomorrow* (pp. 1–6). <http://dx.doi.org/10.1109/RTSI.2016.7740576>.
- Ong, Z. Y., Saidatul, A., & Ibrahim, Z. (2018). Power spectral density analysis for human EEG-based biometric identification. In *2018 international conference on computational approach in smart systems design and applications* (pp. 1–6). IEEE.
- Pippa, E., Zacharakis, E. I., Koutroumanidis, M., & Megalooikonomou, V. (2017). Data fusion for paroxysmal events' classification from EEG. *Journal of Neuroscience Methods*, 275, 55–65. <http://dx.doi.org/10.1016/j.jneumeth.2016.10.004>.
- Pippa, E., Zacharakis, E. I., Mporas, I., Tsirka, V., Richardson, M. P., Koutroumanidis, M., et al. (2016). Improving classification of epileptic and non-epileptic EEG events by feature selection. *Neurocomputing*, 171, 576–585. <http://dx.doi.org/10.1016/j.neucom.2015.06.071>.
- Rasheed, K., Qayyum, A., Qadir, J., Sivathamboo, S., Kwan, P., Kuhlmann, L., et al. (2021). Machine learning for predicting epileptic seizures using EEG signals: A review. *IEEE Reviews in Biomedical Engineering*, 14, 139–155.
- Rosenow, F., Klein, K. M., & Hamer, H. M. (2015). Non-invasive EEG evaluation in epilepsy diagnosis. *Expert Review of Neurotherapeutics*, 15(4), 425–444.
- Sánchez Bonilla, S. (2024). *Enhancing graph neural network performance for early epilepsy detection via EEG signal analysis* (Master's thesis), Universitat Politècnica de Catalunya.
- Selvam, V. S., & Devi, S. S. (2015). Analysis of spectral features of EEG signal in brain tumor condition. *Measurement Science Review*, 15(4), 219–225.
- Shi, Y., Huang, Z., Wang, W., Zhong, H., Feng, S., & Sun, Y. (2020). Masked label prediction: Unified message passing model for semi-supervised classification. CoRR abs/2009.03509, arXiv:2009.03509, URL <https://arxiv.org/abs/2009.03509>.
- Stam, C. J., Nolte, G., & Daffertshofer, A. (2007). Phase lag index: Assessment of functional connectivity from multi channel EEG and MEG with diminished bias from common sources. *Human Brain Mapping*, 28(11), 1178–1193.
- Tatum, W., Rubboli, G., Kaplan, P., Mirsafari, S., Radhakrishnan, K., Gloss, D., et al. (2018). Clinical utility of EEG in diagnosing and monitoring epilepsy in adults. *Clinical Neurophysiology*, 129(5), 1056–1082.
- Umesh, S., Tikka, S. K., Goyal, N., Sinha, V. K., & Nizamie, S. H. (2017). Aberrant gamma band cortical sources and functional connectivity in adolescents with psychogenic non-epileptic seizures: A preliminary report. *Psychiatry Research*, 247, 51–54.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., et al. (2017). Attention is all you need. In I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, R. Garnett (Eds.), *Advances in neural information processing systems: vol. 30*, Curran Associates, Inc.
- Veličković, P., Cucurull, G., Casanova, A., Romero, A., Liò, P., & Bengio, Y. (2018). Graph attention networks. arXiv:1710.10903.
- Vialatte, F., Cichocki, A., Dreyfus, G., Musha, T., Shishkin, S. L., & Gervais, R. (2005). Early detection of Alzheimer's disease by blind source separation, time frequency representation, and bump modeling of EEG signals. In *Artificial neural networks: biological inspirations-ICANN 2005: 15th international conference, Warsaw, Poland, September 11-15, 2005. Proceedings, Part I 15* (pp. 683–692). Springer Berlin Heidelberg.
- Wang, J., Liang, S., He, D., Wang, Y., Wu, Y., & Zhang, Y. (2020). A sequential graph convolutional network with frequency-domain complex network of EEG signals for epilepsy detection. In *2020 IEEE international conference on bioinformatics and biomedicine* (pp. 785–792). IEEE.
- Wu, Z., Pan, S., Chen, F., Long, G., Zhang, C., & Philip, S. Y. (2020). A comprehensive survey on graph neural networks. *IEEE Transactions on Neural Networks and Learning Systems*, 32(1), 4–24.
- Wu, S., Sun, F., Zhang, W., Xie, X., & Cui, B. (2022). Graph neural networks in recommender systems: A survey. *ACM Computing Surveys*, 55(5), 1–37.
- Xu, K., Hu, W., Leskovec, J., & Jegelka, S. (2019). How powerful are graph neural networks? In *International conference on learning representations*.
- Ying, Z., Bourgeois, D., You, J., Zitnik, M., & Leskovec, J. (2019). GNNExplainer: Generating explanations for graph neural networks. In *Advances in neural information processing systems: vol. 32*.
- Yuan, Q., Zhou, W., Li, S., & Cai, D. (2011). Epileptic EEG classification based on extreme learning machine and nonlinear features. *Epilepsy Research*, 96(1), 29–38. <http://dx.doi.org/10.1016/j.eplepsyres.2011.04.013>.
- Zhao, Y., Dong, C., Zhang, G., Wang, Y., Chen, X., Jia, W., et al. (2021). EEG-based seizure detection using linear graph convolution network with focal loss. *Computer Methods and Programs in Biomedicine*, 208, Article 106277. <http://dx.doi.org/10.1016/j.cmpb.2021.106277>.
- Zhao, Y., & He, L. (2015). Deep learning in the EEG diagnosis of Alzheimer's disease. In C. Jawahar, & S. Shan (Eds.), *Computer vision - ACCV 2014 workshops* (pp. 340–353). Cham: Springer International Publishing.
- Zhou, J., Cui, G., Hu, S., Zhang, Z., Yang, C., Liu, Z., et al. (2020). Graph neural networks: A review of methods and applications. *AI Open*, 1, 57–81. <http://dx.doi.org/10.1016/j.aiopen.2021.01.001>.
- Zsom, A., Tsekhan, S., Hamid, T., Levin, J., Truccolo, W., LaFrance, W. C., et al. (2019). Ictal autonomic activity recorded via wearable-sensors plus machine learning can discriminate epileptic and psychogenic nonepileptic seizures. In *Annu int conf IEEE eng med biol soc: vol. 2019*, (pp. 3502–3506). United States.